

- What is a circle packing?
 - definition
 - complex and border circles $\rightarrow$ packing
 - * complex + circles = geometry.
 - complex and max_pack $\rightarrow$ packing on sphere
 - mobius map/projections between geometries
 - * KAT theorem.
 - * Local rigidity with global flexibility.
 - * "Circle packings are interesting because they provide a canonical and conformally natural way to embed a planar graph into a surface. " -Rhode
- hierarchy
- flowers -monotonicity
 - proof in D
- Existence theorem on disc.
 - gives sphere.
 - automorphisms are different Möbius transformations of same CP.
- Uniqueness
- DAF
 - computation/analogue/approximation
 - Riemann mapping theorem (use conformal mapping I script).
 - * Key: started field.
 - * Different construction methods.
 - * refinement gives classic map and proves result.
 - * List/show? other results
- Branching
 - classic example with branching
 - basic example of non- 2π flowers.
 - * polynomial branching.
 - * discretization effects.
- fractional branching

- coherence/holonomy
 - computation/experimentation
 - tri-branch property
 - existence
 - uniqueness? -special case
 - sphere realization -loss of monotonicity.
- Other application
 - probability/Brownian motion
 - Brain mapping
- CirclePack!
- Research opportunities
- Bib